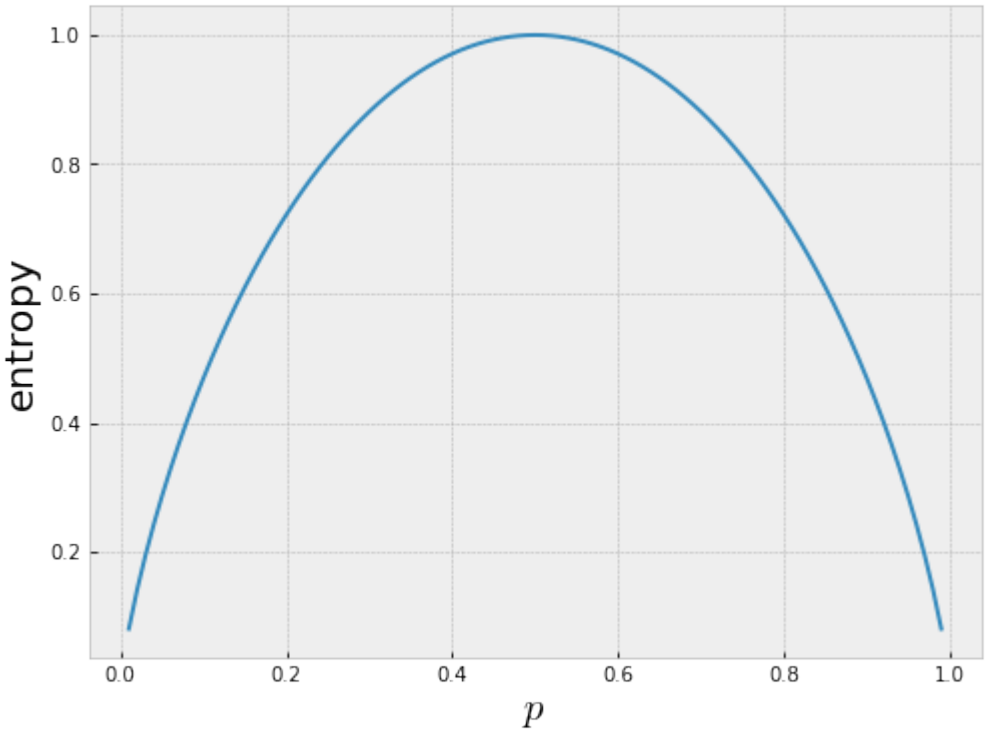


Figure 6.1 Entropy of a Bernoulli random variable as a function of the success probability. The entropy is maximized (uncertainty is at the highest) when the success probability is at 0.5.

Both insights are valuable when we study our uncertainty about whether the coin will land on heads or not. First, we shouldn't have negative uncertainty about something, so it makes sense that the entropy is never negative. More importantly, entropy is maximized right at the middle when $p = 0.5$. This is quite reasonable: as p gets further and further away from 0.5, we will become more certain about the outcome of the event, whether the coin will land on heads or not.

For instance, if $p = 0.7$, then we are more certain that it will land on heads—our entropy is around 0.9 here. If $p = 0.1$, then we are even more certain about the outcome (this time, that it will land on tails)—the entropy here is roughly 0.5. While entropy is not defined at the endpoints (due to the logarithmic function), entropy approaches zero as we get closer to either endpoint, indicating zero uncertainty. When $p = 0.5$, on the other hand, our uncertainty is at its maximum, as we are maximally unsure whether the coin will land on heads or on tails. These calculations demonstrate that entropy is an appropriate measure of uncertainty.

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